

Transmission



Special Tool(s)

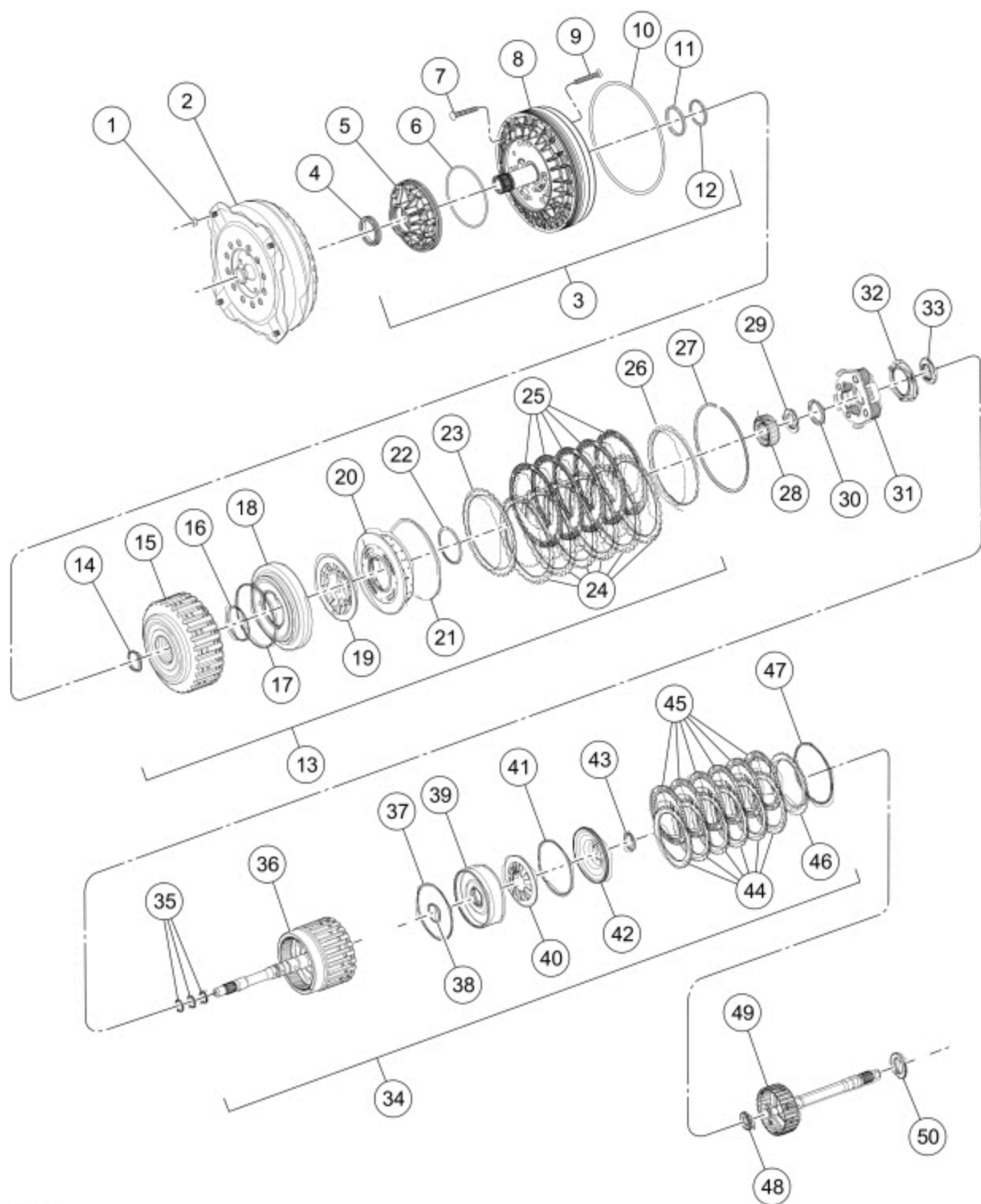
 ST1214-A	Dial Indicator Gauge with Holding Fixture 100-002 (TOOL-4201-C)
 ST1360-A	Drawbar (Heavy Duty Threaded) 204-029 (T77F-1176)
 ST2683-A	End Play Gauge, Transmission 307-534
 ST1631-A	Handle, Torque Converter 307-091 (T81P-7902-C)
 ST3059-A	Installer, 6R80 Rear Seal 4X4 307-637
 ST3060-A	Installer, 6R80 Rear Seal 4X2 307-638
 ST2887-A	Installer, Fluid Pump Seal 307-556

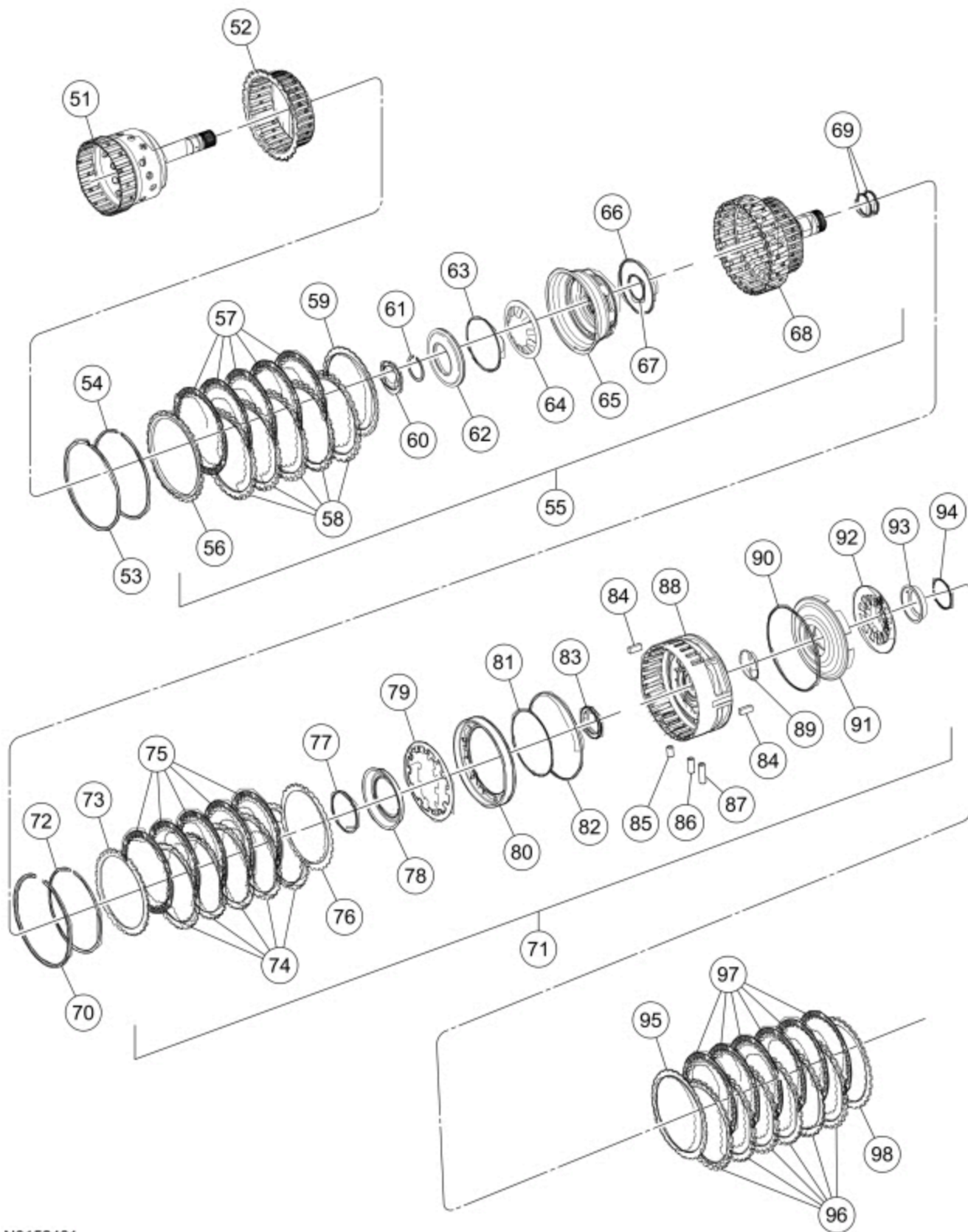
	Installer, Front Wheel Hub Oil Seal 205-276 (T88T-1175-AH)
 ST2890-A	Installer, Shifter Fluid Seal 307-559
 ST2894-A	Output Shaft Bearing Installer 307-647
 ST3061-A	Remover/Installer, 6R80 Rear Bearing 307-639
 ST2891-A	Remover, Transmission Fluid Pump 307-553
 ST1104-B	Retaining Ring Pliers 307-343 (T95P-77001-AHR)

Material

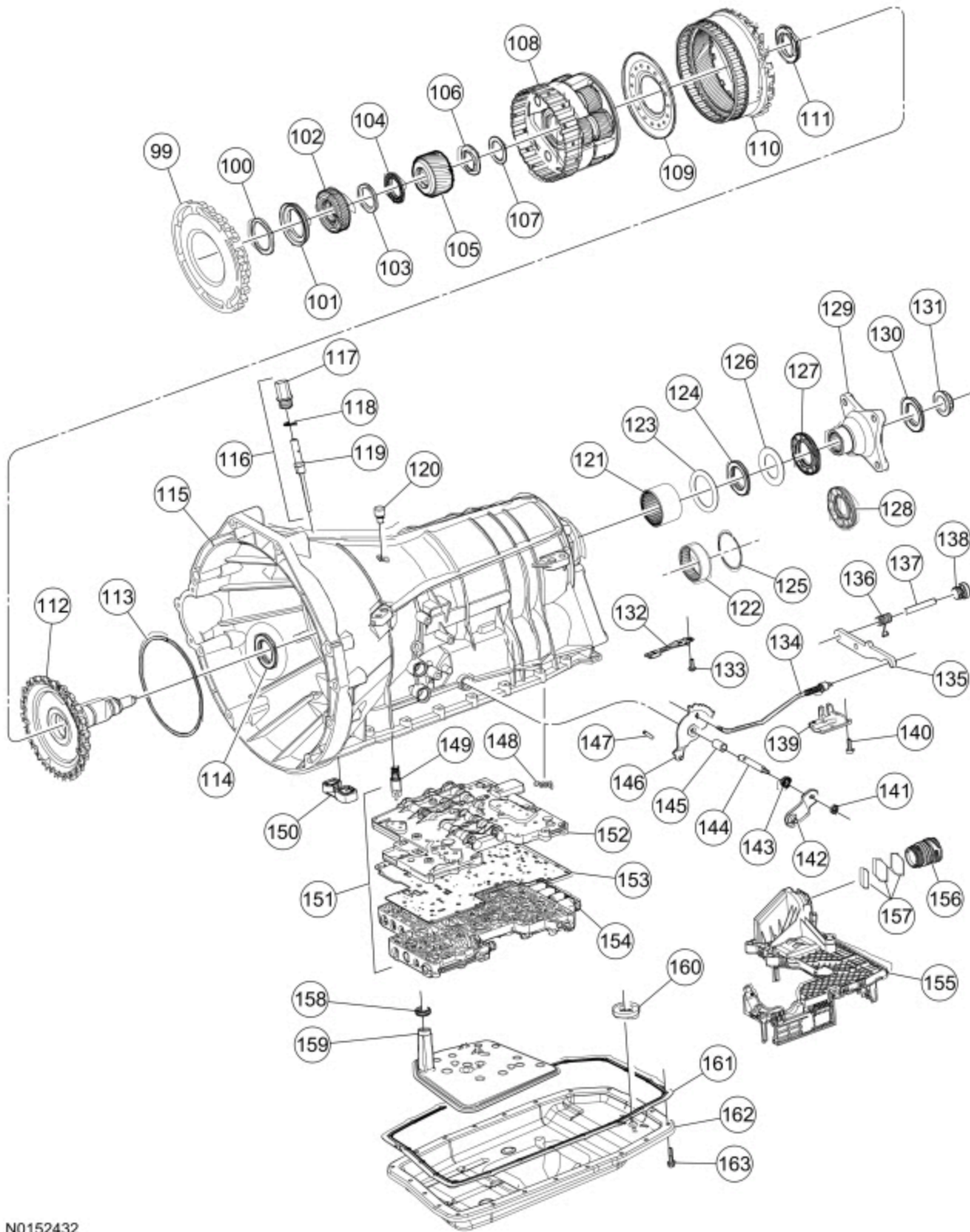
Item	Specification
Motorcraft® MERCON® LV Automatic Transmission Fluid XT-10-QLVC (US); CXT-10-LV12 (Canada)	MERCON® LV

Disassembled Views





N0152431



N0152432

Item	Part Number	Description
1	N800750 N800750 -S437-S437	Flexplate-to-torque converter nut (4 required)
2	7902 7902	Torque converter
3	7A103 7A103	Pump assembly
4	7A248 7A248	Front pump oil seal
5	—	Pump body (part of 7A103)
6	7A248 7A248	Front pump inner oil seal
7	7N134 7N134	Front pump-to-case bolt (13 required)

8	7A103 7A103	Front pump assembly
9	W707871 W707871 -S300-S300	Bolt — attaches pump to pump adapter assembly (11 required)
10	7A248 7A248	Front pump outer oil seal
11	7D019 7D019	Clutch support fluid seal
12	7L323 7L323	Front pump support seal
13	—	Forward (A) clutch assembly
14	7D014 7D014	Front pump selective washer
15	7A360 7A360	Forward (A) clutch cylinder and hub assembly
16	7A548 7A548	Forward (A) clutch inner piston seal
17	7A548 7A548	Forward (A) clutch outer piston seal
18	7A262 7A262	Forward (A) clutch piston
19	7A480 7A480	Forward (A) clutch piston retaining spring
20	7H360 7H360	Forward (A) clutch balance piston assembly
21	7A548 7A548	Forward (A) clutch balance piston outer seal
22	7H365 7H365	Forward (A) clutch balance piston snap ring
23	7E085 7E085	Forward (A) clutch cushion spring
24	7E314 7E314	Forward (A) clutch steel plates (externally splined) (quantity model dependent)
25	7B164 7B164	Forward (A) clutch friction plates (internally splined) (quantity model dependent)
26	7B066 7B066	Forward (A) clutch pressure plate
27	7D483 7D483	Pressure plate retaining snap ring
28	7D063 7D063	Front planetary sun gear (No. 1)
29	7H375 7H375	Bearing (T1)
30	—	Front planetary carrier assembly snap ring (part of 7A398)
31	7A398 7A398	Front planetary carrier assembly
32	7L339 7L339	Plate transmission fluid collector
33	7L495 7L495	Bearing (T2)
34	—	Overdrive (E) clutch assembly
35	7G091 7G091	Turbine shaft seals (3 required)
36	7F207 7F207	Input shaft
37	7A548 7A548	Overdrive (E) clutch piston outer seal
38	7A548 7A548	Overdrive (E) clutch piston inner seal
39	7A262 7A262	Overdrive (E) clutch piston
40	7B070 7B070	Overdrive clutch piston spring
41	7A548 7A548	Balance piston outer seal
42	7H360 7H360	Overdrive (E) balance piston
43	7C122 7C122	Balance piston snap ring
44	7B442 7B442	Overdrive (E) clutch steel plates (externally splined)

45	7B164 7B164	Overdrive (E) clutch friction plates (internally splined)
46	7B066 7B066	Overdrive (E) pressure plate
47	7D483 7D483	Overdrive (E) clutch retaining ring
48	7C096 7C096	Bearing (T3)
49	7F351 7F351	Intermediate (C) clutch shaft
50	7H375 7H375	Bearing (T4)
51	7B067 7B067	Sun gear hub and shaft assembly
52	7F236 7F236	Direct (B) clutch hub
53	7C122 7C122	Direct (B) clutch cylinder retaining ring
54	7D483 7D483	Direct (B) clutch pressure plate retaining ring
55	—	Direct (B) clutch assembly (clutch B)
56	7B066 7B066	Direct (B) clutch pressure plate
57	7B164 7B164	Direct (B) clutch friction plates (internally splined) (quantity model dependent)
58	7B442 7B442	Direct (B) clutch steel plates (externally splined) (quantity model dependent)
59	7E085 7E085	Direct (B) clutch cushion plate
60	7C096 7C096	Roller bearing (T5)
61	7A577 7A577	Direct (B) clutch piston retaining ring
62	7A262 7A262	Direct (B) clutch balance piston
63	7A548 7A548	Direct (B) clutch balance seal
64	7B488 7B488	Direct (B) clutch piston return spring
65	—	Direct (B) clutch piston
66	7A548 7A548	Direct (B) clutch piston outer seal
67	7C099 7C099	Direct (B) clutch piston inner seal
68	7F283 7F283	Direct (B) clutch cylinder
69	7D020 7D020	Shell cylinder seals (2 required)
70	7D483 7D483	Center support retaining ring
71	—	Intermediate (C) clutch assembly
72	7D483 7D483	Intermediate (C) clutch pressure plate retaining ring
73	7B066 7B066	Intermediate (C) clutch pressure plate
74	7B442 7B442	Intermediate (C) clutch steel plates (externally splined) (quantity model dependent)
75	7B164 7B164	Intermediate (C) clutch friction plates (internally splined) (quantity model dependent)
76	7E085 7E085	Intermediate (C) clutch pressure plate spring
77	7A577 7A577	Intermediate (C) clutch retaining ring
78	7B043 7B043	Intermediate (C) clutch ring
79	7A480 7A480	Intermediate (C) clutch piston spring
80	7E005 7E005	Intermediate (C) clutch piston
81	7F225 7F225	Intermediate (C) clutch piston inner seal

82	7F224 7F224	Intermediate (C) clutch piston outer seal
83	7F373 7F373	Bearing (T6)
84	7B220 7B220	Center support keys (2 required)
85	7G199 7G199	Center support feed tube seal (black)- early build 2 required - late build 1 required
86	7G087 7G087	Center support feed tube seal (green)
87	7G084 7G084	Center support feed tube seal (blue)
88	7G033 7G033	Center support assembly
89	7D404 7D404	Low/reverse clutch piston center seal
90	7D403 7D403	Low/reverse clutch piston outer seal
91	7A262 7A262	Low/reverse clutch piston
92	7B070 7B070	Low/reverse clutch piston return spring
93	7H318 7H318	Low/reverse clutch piston retainer
94	7D483 7D483	Retainer transmission-snap ring
95	7E085 7E085	Clutch disc cushion plate
96	7B442 7B442	Low/reverse clutch steel plates (externally splined)
97	7B164 7B164	Low/reverse clutch friction plates (internally splined)
98	7B066 7B066	Low/reverse clutch pressure plate (select fit)
99	7A089 7A089	Q.W.C
100	7F405 7F405	Thrust gear shim (select fit)
101	7C041 7C041	Thrust bearing (T7)
102	7D063 7D063	Sun gear No. 2
103	7D235 7D235	Thrust bearing outer race
104	7D234 7D234	Roller bearing (T8)
105	7D063 7D063	Sun gear No. 3
106	7D234 7D234	Roller bearing (T9)
107	—	Roller bearing race (T9)
108	7D006 7D006	Rear planetary carrier assembly
109	—	Fluid collar rear planetary plate (part of 7D006)
110	7A153 7A153	Output shaft ring gear assembly
111	7G178 7G178	Thrust Bearing (T10)
112	7060 7060	Output shaft park gear assembly
113	7N194 7N194	Output shaft retaining ring
114	7B368 7B368	Bearing (T11)
115	7005 7005	Transmission case assembly (model dependent)
116	7A0107 7A010	Transmission case fluid fill plug assembly
117	7H398 7H398	Transmission case fluid fill plug
118	7A0107 7A010	Transmission case fluid fill seal
119	7A0107 7A010	Transmission oil level indicator

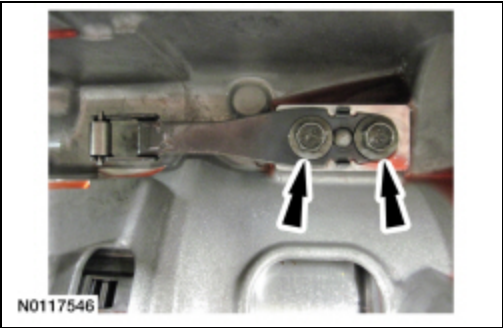
120	7034 7034	Transmission case vent assembly (model dependent)
121	7A415 7A415	Output shaft bearing assembly (R.W.D.)
122	7A415 7A415	Output shaft bearing assembly (4.W.D.)
123	7A433 7A433	Washer (R.W.D.)
124	7B368 7B368	Thrust bearing (T12) (R.W.D.)
125	7030 7030	Output shaft bearing snap ring (4.W.D.)
126	7N357 7N357	Slip plane washer (R.W.D.)
127	7052 7052	Output shaft seal (R.W.D.)
128	7052 7052	Output shaft seal (4.W.D.)
129	7089 7089	Output shaft flange (R.W.D.)
130	7052 7052	Output shaft flange seal (R.W.D.)
131	7045 7045	Output shaft flange retaining nut (R.W.D.)
132	7E332 7E332	Manual valve detent spring
133	W711235 W711235 -S300-S300	Manual valve detent spring retaining screw and washer
134	7A232 7A232	Park pawl actuator rod
135	7A441 7A441	Park pawl
136	7D070 7D070	Park pawl return spring
137	7D071 7D071	Park pawl shaft
138	7H398 7H398	Plug assembly transmission case housing
139	7G101 7G101	Park rod actuating plate
140	W711235 W711235 -S300-S300	Park pawl abutment retaining screw and washer
141	W708455 W708455 -S441-S441	Manual control lever nut
142	7A256 7A256	Manual control lever
143	7B498 7B498	Manual control lever shaft seal
144	7C493 7C493	Manual control lever shaft
145	7A209 7A209	Manual control lever shaft spacer
146	7A115 7A115	Manual valve detent lever assembly
147	7G100 7G100	Manual valve detent lever retaining pin
148	7F277 7F277	Q.W.C. bias spring
149	7H322 7H322	Thermal bypass valve
150	7F401 7F401	Front pump adapter seal
151	7A100 7A100	Main control assembly
152	—	Main control valve body assembly (part of 7A100)
153	7Z490 7Z490	Main control valve body separator plate
154	—	Lower main control valve body (part of 7A100)
155	7G276 7G276	Molded leadframe
156	7G276 7G276	Bulkhead connector sleeve

157	7A248 7A248	Bulkhead connector sleeve seals
158	—	Transmission fluid filter seal (part of 7A098)
159	7A098 7A098	Transmission fluid filter
160	—	Transmission fluid pan magnet
161	7A191 7A191	Transmission fluid pan gasket
162	7A194 7A194	Transmission fluid pan
163	W500214 W500214 -S437-S437	Transmission fluid pan bolt

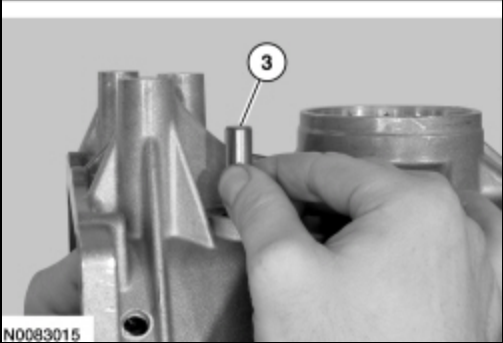
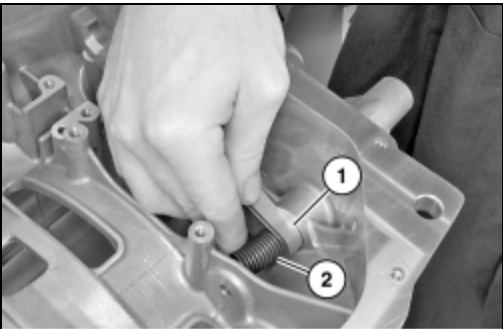
NOTE: Clutch plate quantity is model dependant based on engine displacement. Refer to Clutch Plate Quantity chart in the Specifications portion of this section.

All vehicles

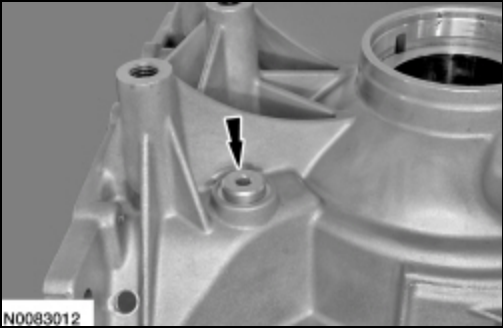
1. Install the park detent spring and the 2 bolts.
 - Tighten to 12 Nm (106 lb-in).



2. Position the park pawl and spring into the case and install the park pawl pin.
 1. Park pawl
 2. Park pawl return spring
 3. Park pawl pin



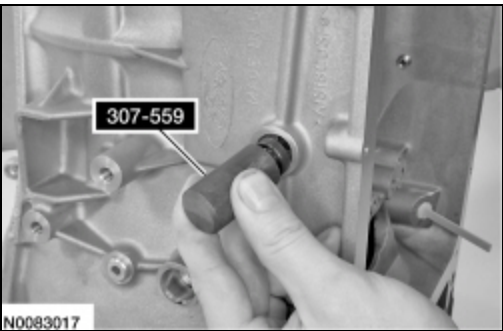
3. Install the park pawl pin bolt.
 - Tighten to 23 Nm (17 lb-ft).



4. Install a new manual control lever shaft seal on the Shifter Fluid Seal Installer.



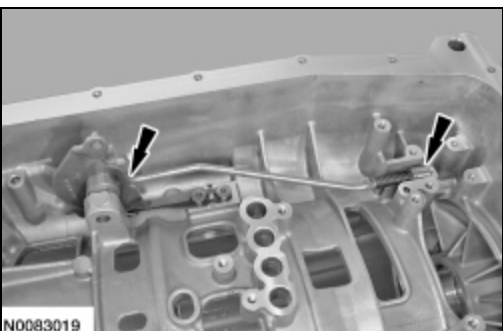
5. Using the Shifter Fluid Seal Installer, install the manual control lever shaft seal.



6. Assemble the manual control lever detent plate assembly and park pawl actuator rod.

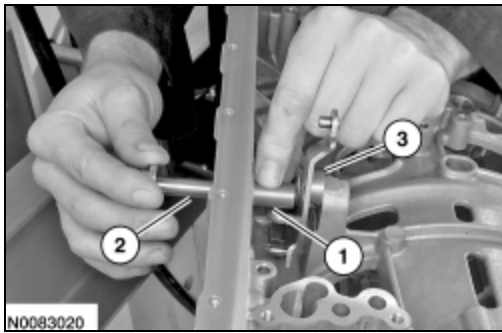


7. Position the manual valve detent lever and park pawl actuator rod assembly in the transmission case.

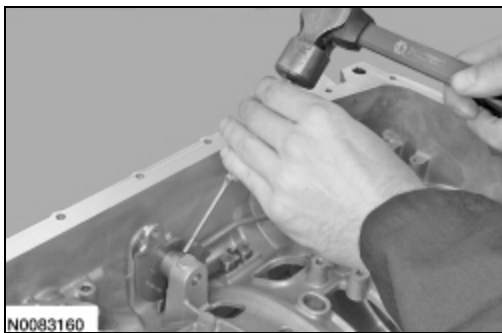


8. Position the manual control lever spacer in place and slide the manual control lever shaft into the case and the manual lever detent plate.

1. Manual control lever spacer
2. Manual control lever shaft
3. Manual control lever detent plate

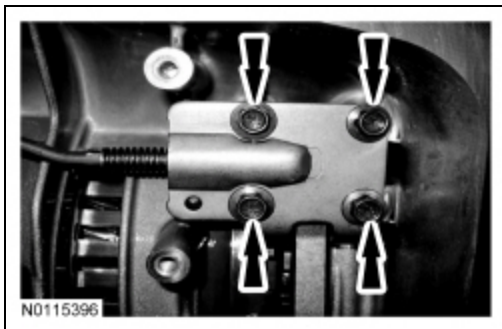


9. Align the roll pin hole in the manual control lever detent plate with the roll pin hole in the manual control lever shaft and install a new roll pin using a suitable tool.



10. Install the park rod actuating plate and the 4 bolts.

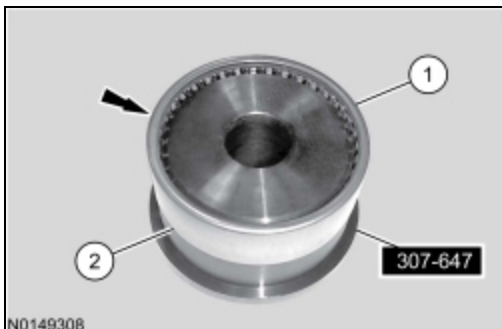
- Tighten to 12 Nm (106 lb-in).



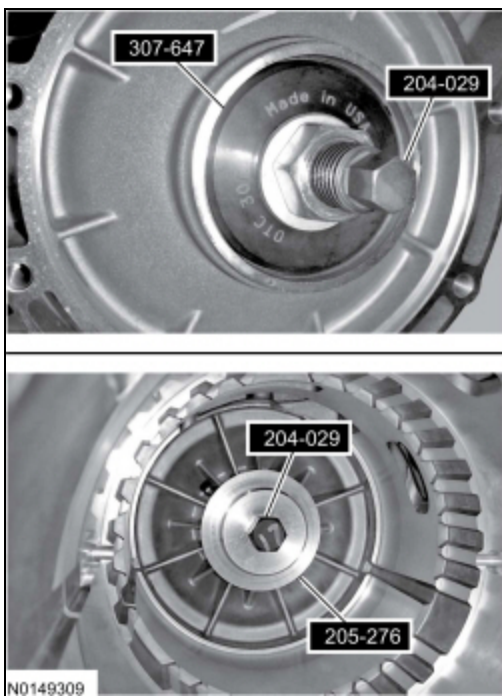
Four-Wheel Drive (4WD) vehicles

11. If removed, position a new bearing on the Output Shaft Bearing Installer. Install the bearing with the flat bearing surface facing down and the rounded bearing surface facing up so that when installed, the flat surface is on the snap ring side and the rounded surface is on the transmission case side.

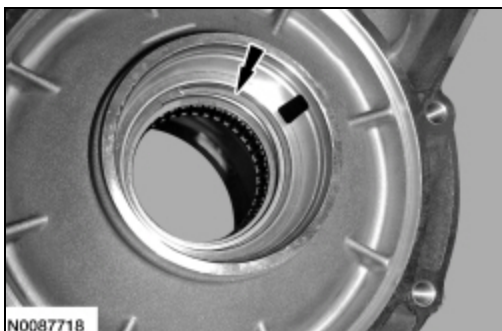
1. Rounded bearing surface facing up
2. Flat bearing surface facing down



12. Using the Output Shaft Bearing Installer, the Drawbar and the Front Wheel Hub Oil Seal Installer, install the bearing in the transmission case.

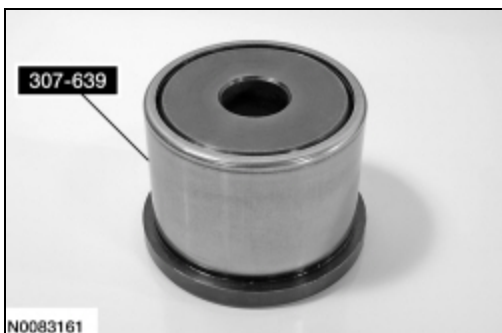


13. Install the snap ring.

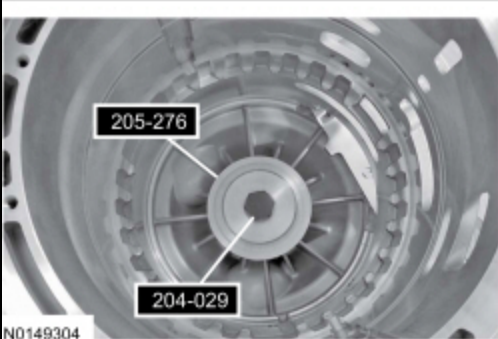
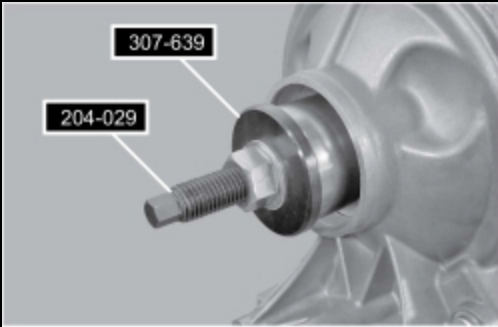


Rear Wheel Drive (RWD) vehicles

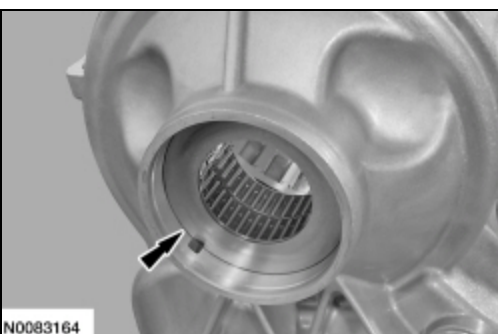
14. Install the bearing on the 6R80 Rear Bearing Remover/Installer.



15. Assemble the Front Wheel Hub Oil Seal Installer and the 6R80 Rear Bearing Remover/Installer on the Drawbar and install the bearing in the transmission case.



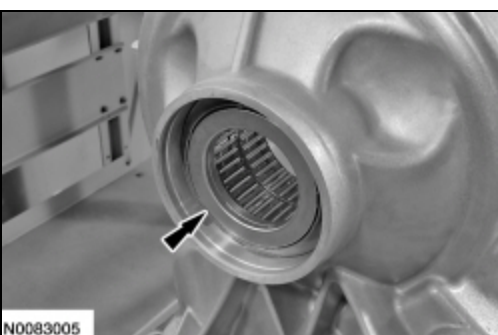
16. Position the thrust bearing spacer.



17. Using the 6R80 Rear Bearing Remover/Installer and a press, press the thrust bearing spacer in place.



18. Install the T12 thrust bearing.

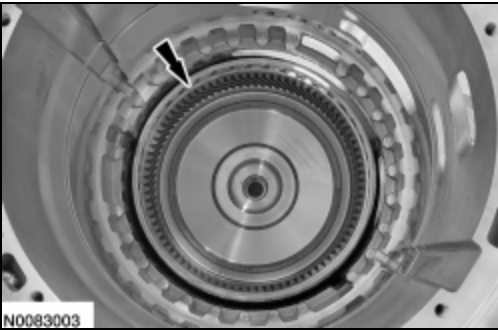


All vehicles

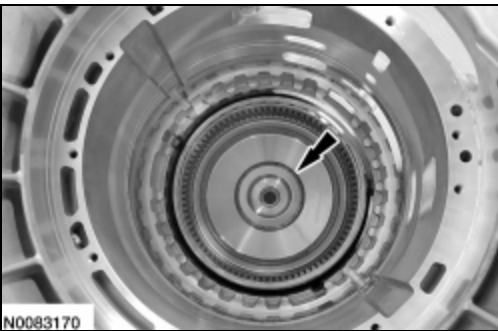
19. Install the T11 bearing, on the planetary ring gear and output shaft assembly.



20. Install the rear planetary ring gear and output shaft assembly.



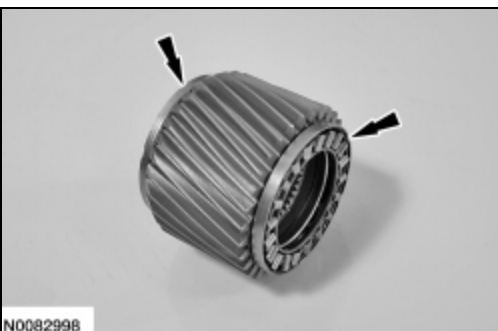
21. Install the T10 thrust bearing on the output shaft assembly.



22. Install the T9 roller bearing race into the bottom of the carrier.

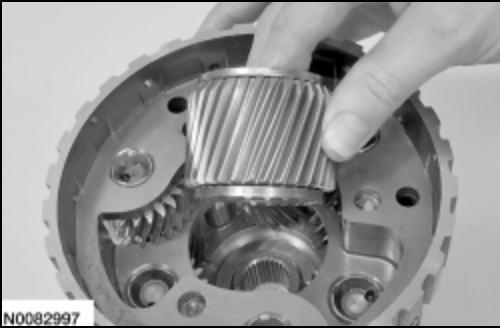


23. Install the top T8 and the bottom T9 roller bearings onto the sun gear.



24. **NOTE:** When installing the sun gear, make sure that the **TAPERED EDGE** is facing **UP** toward the torque converter housing.

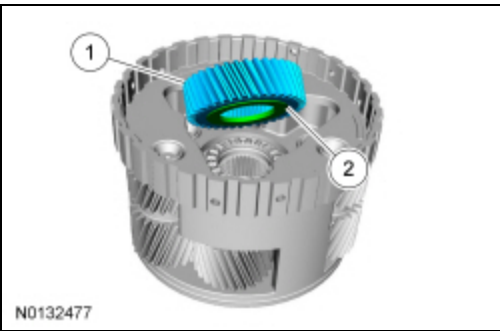
Install the sun gear into the planetary assembly.



25. **NOTE:** Be sure the bearing race is installed in the sun gear.

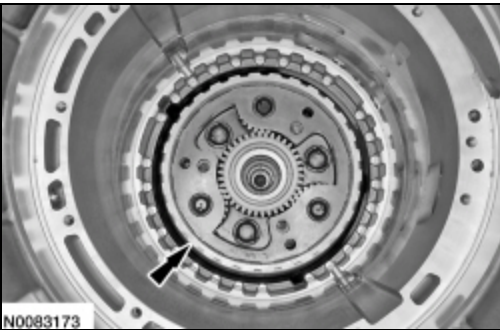
Install the low/reverse sun gear and bearing race into the planetary carrier with the race side facing down.

1. Low/reverse sun gear
2. Bearing race



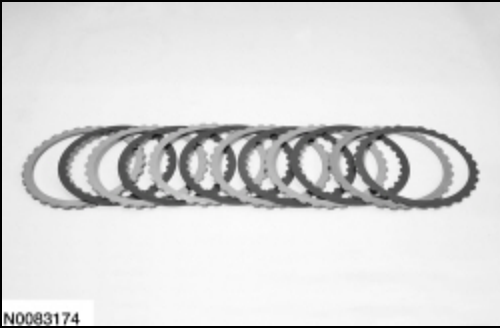
26. **NOTE:** Place the manual control lever in **PARK** when installing the rear planetary gearset to hold the rear ring gear stationary.

Install the planetary carrier.

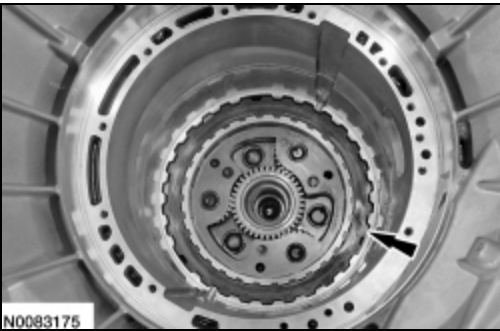


27. **NOTE:** Friction and steel plate quantity are model dependent and vary based on engine displacement. For additional information, refer to the *Clutch Plate Quantity Chart* in the *Specifications* portion of this section.

Soak the low/reverse clutch plates in clean automatic transmission fluid.

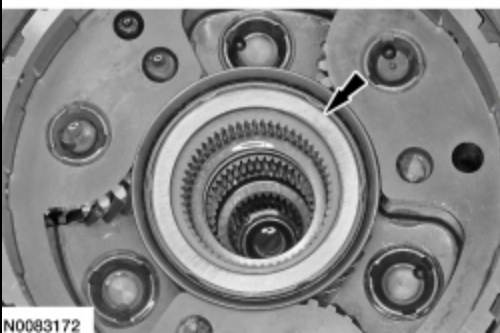
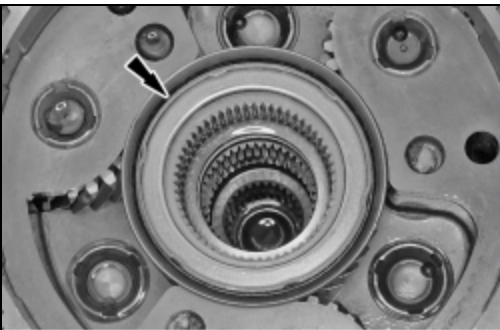


28. Install the low/reverse clutch plates, starting with the pressure plate and alternating between the friction and steel plates and ending with the wave spring plate on the top.



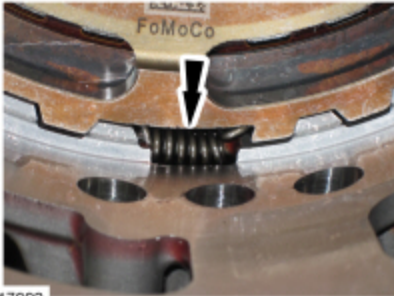
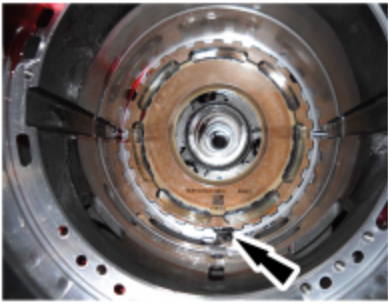
29. **NOTE:** *Inspect the T7 thrust bearing and selective shim for damage or scoring.*

Install the T7 thrust bearing and the selective shim.

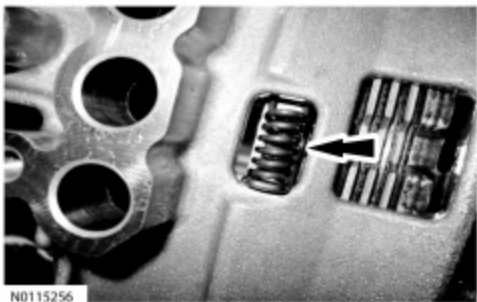


30. **NOTE:** *Align the QWC in the transmission case so the bias spring fits into the transmission case.*

Install the QWC as shown so the bias spring can be installed as shown.



31. Install the bias spring.



32. **NOTE:** Make sure that when installing the center support the feed holes on the center support are lined up with the feed holes in the case.

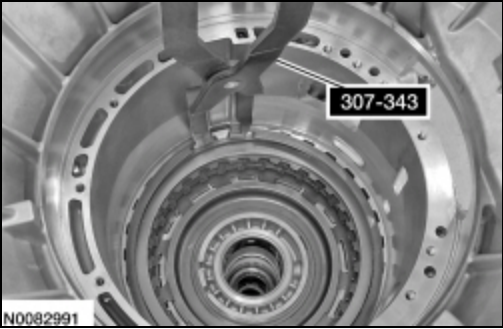
Install the center support into the case.



33. **NOTICE:** Install the snap ring with the beveled surface facing up and the snap ring gap in the 3 o'clock or the 9 o'clock position.

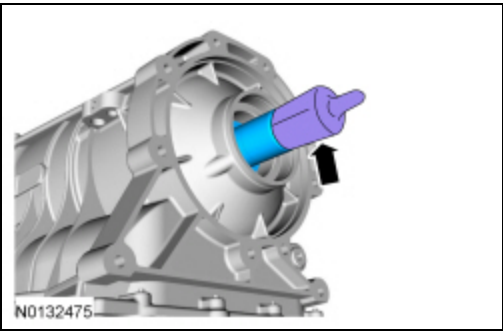
NOTE: Rotate the case to a horizontal position.

Using the Retaining Ring Pliers, install the center support snap ring. Using a suitable tool, make sure the snap ring is fully seated.

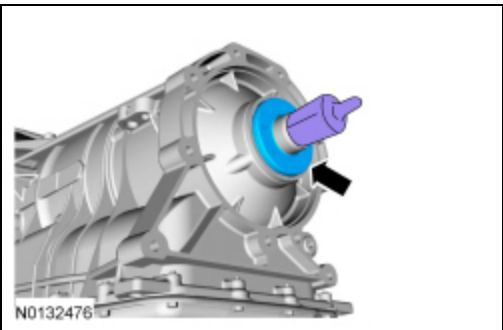


Four-Wheel Drive (4WD) vehicles

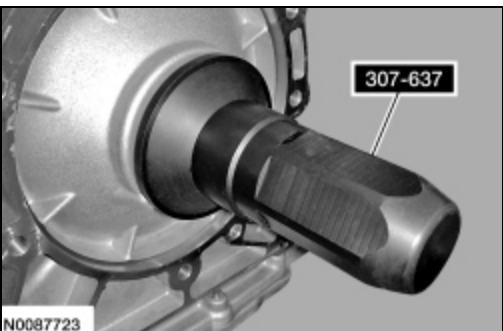
34. Install the seal protector provided with the new output shaft seal on the output shaft.



35. Slide the output shaft seal over the output shaft seal protector onto the output shaft. Remove the output shaft seal protector.



36. Using the 6R80 Rear Seal 4X4 Installer, install the output shaft seal.



Rear Wheel Drive (RWD) vehicles

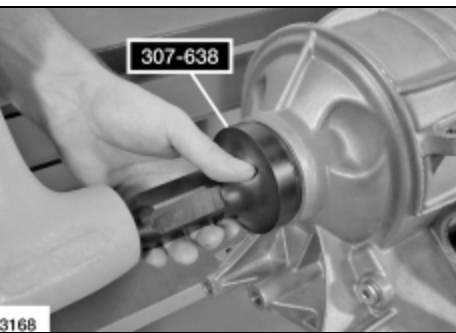
37. Rotate the transmission to the vertical position and install the slip plane washer.



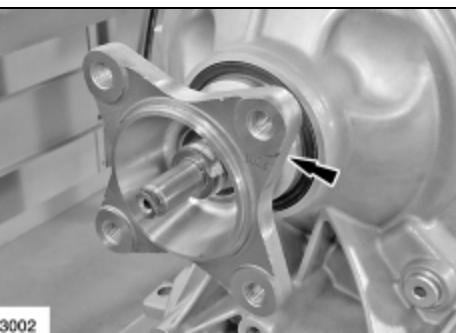
38. Position a new output shaft seal on the 6R80 4X2 Rear Seal Installer.



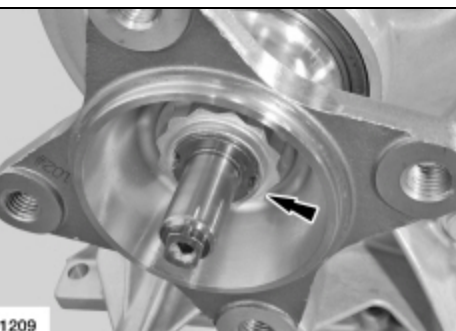
39. Using the 6R80 Rear Seal 4X2 Installer, install the output shaft seal.



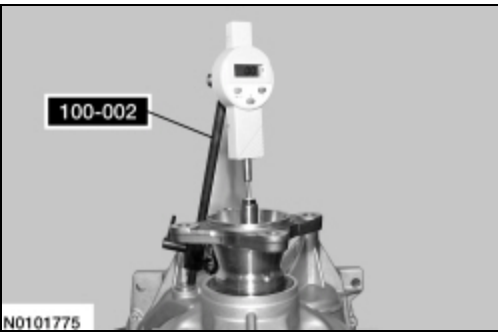
40. Install the output shaft flange and the output shaft flange seal.



41. Loosely install a new output shaft flange retaining nut.



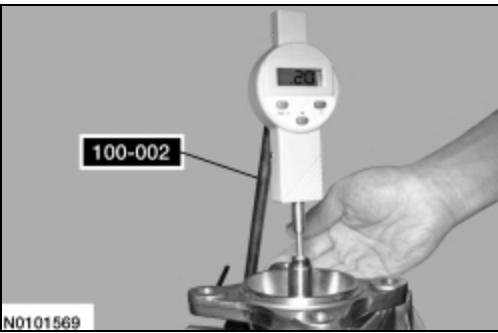
42. Install the Dial Indicator Gauge with Holding Fixture on the transmission case and position the plunger on the output shaft flange.



43. **NOTE:** *Loosely install the flange nut or an inaccurate reading may be recorded.*

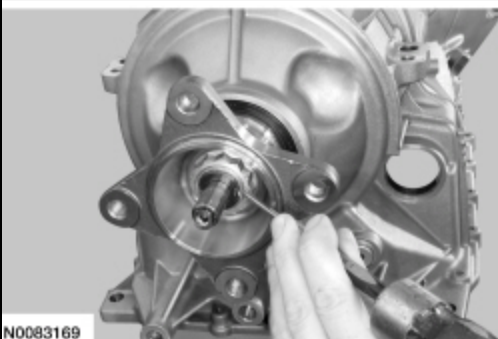
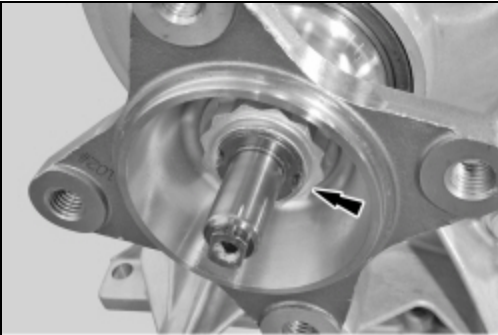
Using the Dial Indicator Gauge with Holding Fixture, lift up on the output shaft flange and record the reading. The reading should be within 0.6-0.9 mm (0.024-0.035 in).

- If the reading is not within specification, install a different (either thinner or thicker) selective shim.



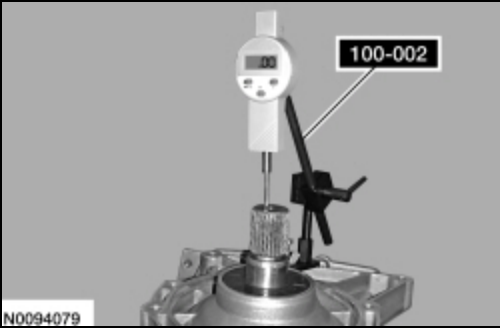
44. Tighten the output shaft flange nut.

- Tighten to 80 Nm (59 lb-ft).
- After installing the new output shaft flange nut, stake the slots to prevent it from coming loose.



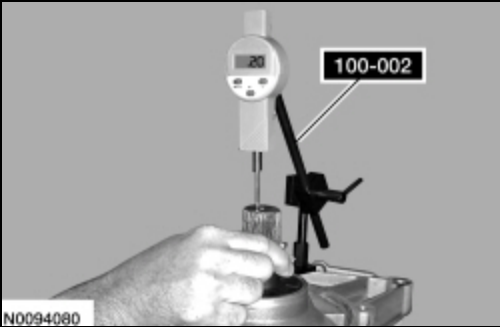
Four-Wheel Drive (4WD) vehicles

45. Install the Dial Indicator Gauge with Holding Fixture on the transmission case and position the plunger on the output shaft.



46. Using the Dial Indicator Gauge with Holding Fixture, lift up on the output shaft and record the reading. The reading should be within 0.15-0.35 mm (0.005-0.013 in).

- If the reading is not within the specification, install a different (either thinner or thicker) select fit thrust gear shim.

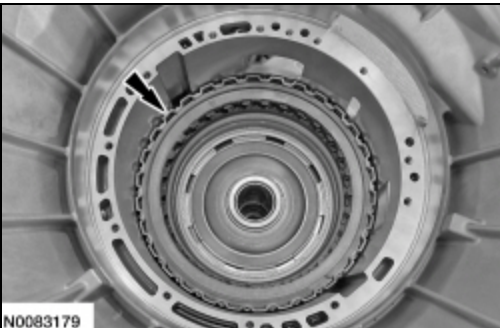


All vehicles

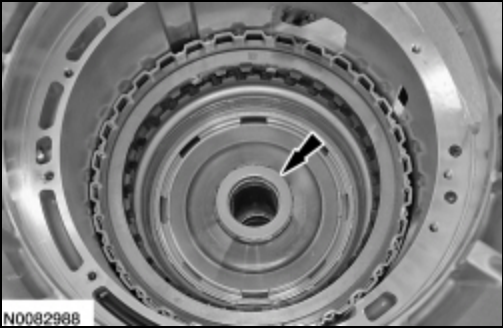
47. Rotate the transmission in the vertical position with the torque converter housing facing up and install the T6 bearing.



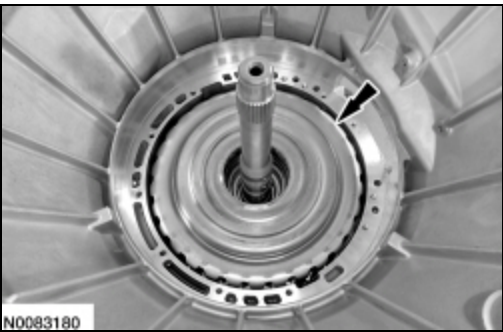
48. Install the direct (B) clutch assembly.



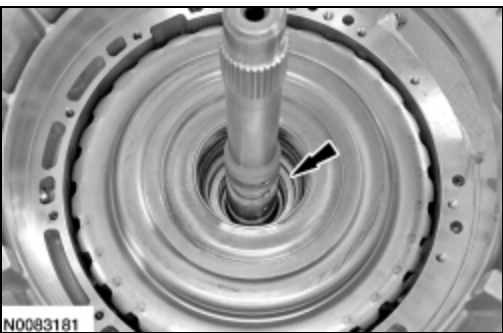
49. Install the T5 roller bearing on the direct (B) clutch assembly.



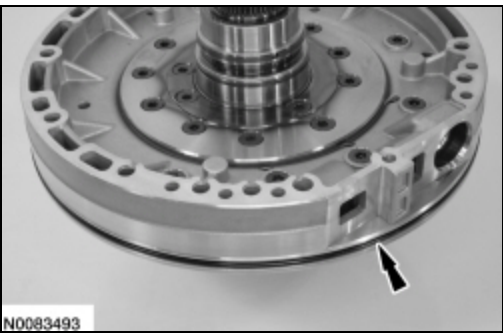
50. Install the forward (A)/overdrive (E) clutch assembly.



51. Install the selective shim.



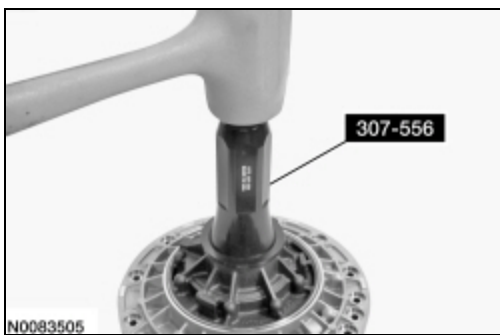
52. If the front pump was not disassembled and assembled, install a new front pump O-ring seal and lubricate it with petroleum jelly.



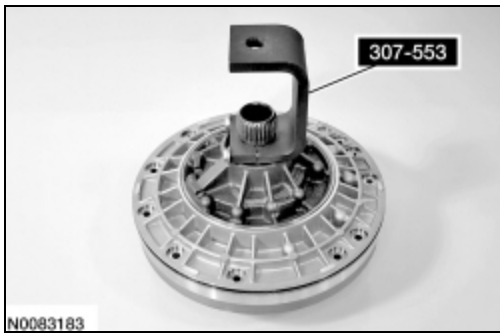
53. Install a new front pump seal on the Fluid Pump Seal Installer.



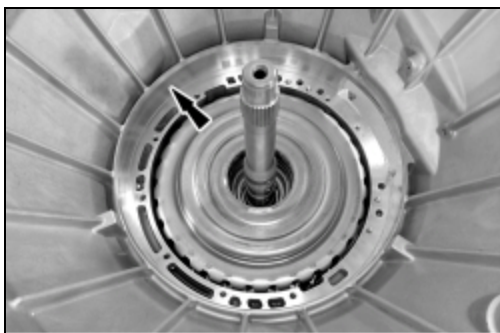
54. Using the Fluid Pump Seal Installer, install the front pump seal.



55. Install the Transmission Fluid Pump Remover on the front pump.



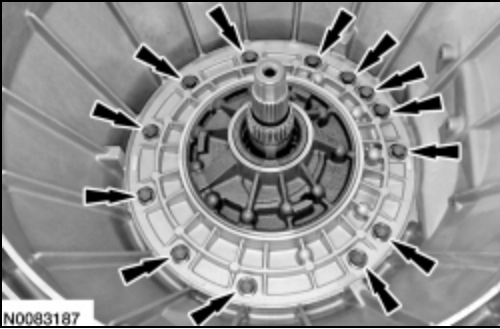
56. Lubricate the transmission case pump bore with petroleum jelly. Using the Transmission Fluid Pump Remover, position the pump assembly in the case and rotate the pump to insert the pump splines into the front planetary sun gear. Rotate the pump to align the pump-to-case bolts and push the pump into the case.



57. **NOTE:** *Failure to replace the fasteners will result in transmission fluid leakage.*

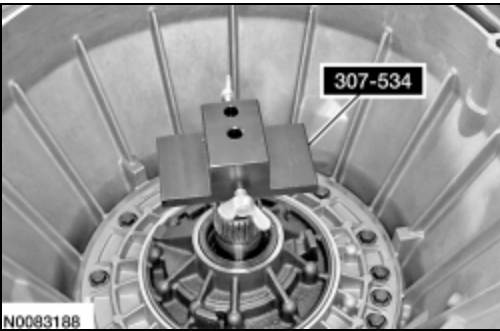
Install 13 new front pump-to-case bolts and washers. Tighten in a crisscross pattern.

- Tighten pump fasteners with washers to 10 Nm (89 lb-in).
- Tighten fasteners without washers to 10 Nm (89 lb-in).



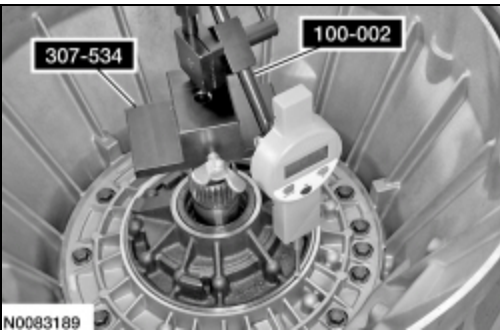
58. Install the Transmission End Play Gauge on the input shaft to measure the end play.

- Tighten the wing nut.



59. Install the Dial Indicator Gauge with Holding Fixture on the Transmission End Play Gauge.

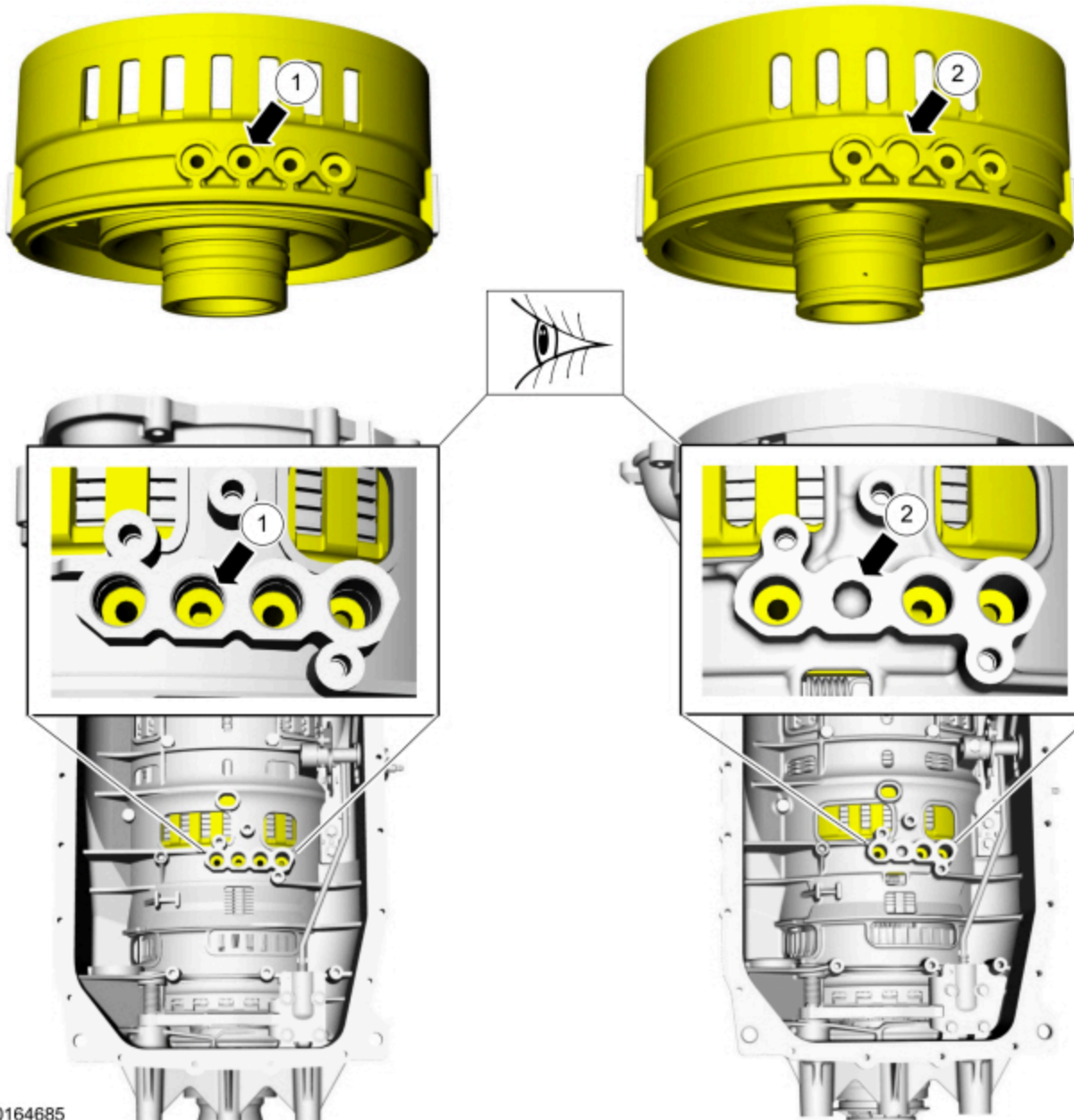
- Push down on the Transmission End Play Gauge and zero the Dial Indicator Gauge.
- Lift up on the Transmission End Play Gauge and record the measurement on the Dial Indicator Gauge. The measurement should be between 0.2-0.4 mm (0.008-0.015 in). If the measurement is not within specification, install a new selective front pump shim.
- Measure the original shim, install a thinner or a thicker shim to achieve the correct measurement.



60. **NOTE:** *Transmissions built with an internal QWC only require 3 center support feed tube seals, as one of the low reverse clutch circuits were eliminated. Transmissions built without a internal QWC with 4 center support fluid passages, require 4 center support feed tube seals.*

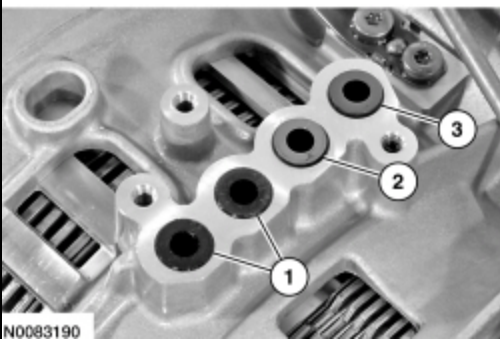
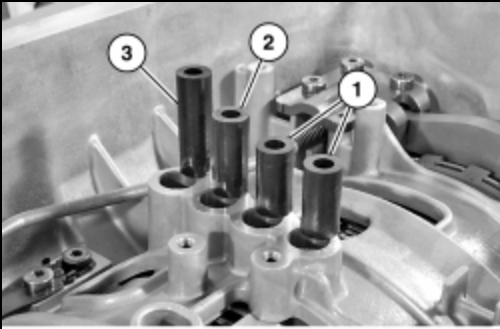
Inspect the the transmission case for the correct number of center support feed tube seals required.

1. Early build transmissions with 4 center support fluid passages.
2. Late build transmissions with 3 center support fluid passages



Vehicles equipped with early build transmissions

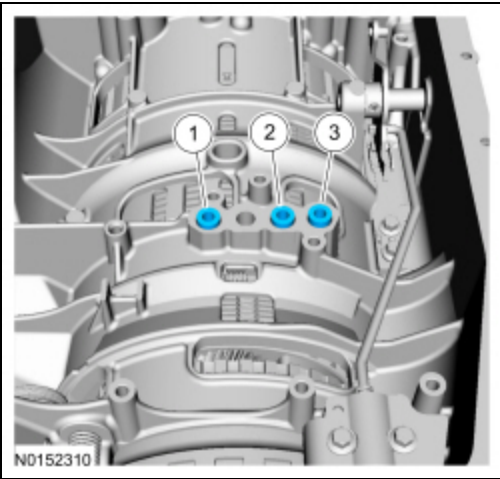
61. Rotate the transmission in the horizontal position with the main control area facing up and install 4 new center support feed tube seals.
- 1. Center support feed tube seal (7G199 Black)
 - 2. Center support feed tube seal (7G087 Green or Orange)
 - 3. Center support feed tube seal (7G084 Blue)



Vehicles equipped with late build transmissions

62. Rotate the transmission in the horizontal position with the main control area facing up and install 3 new center support feed tube seals.

- 1. Center support feed tube seal (7G199 Black)
- 2. Center support feed tube seal (7G087 Green or Orange)
- 3. Center support feed tube seal (7G084 Blue)

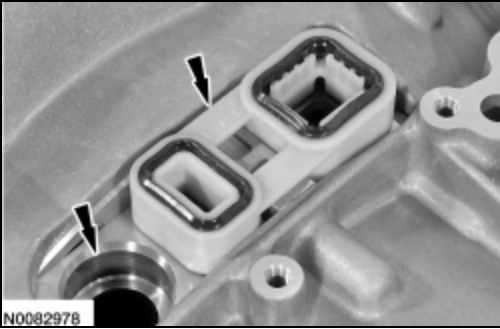


All vehicles

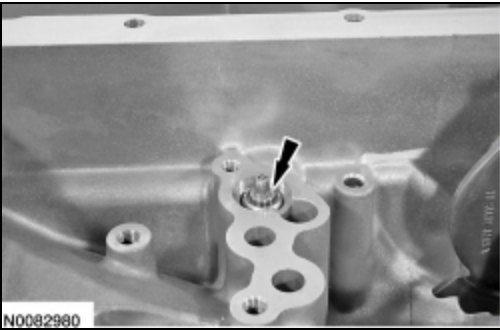
63. **NOTE:** Make sure the transmission fluid filter seal has been removed. If it has not, use a suitable pick and remove the seal.

NOTE: Add 0.118L (4 oz) of clean transmission fluid to prime the pump.

Install the front pump adapter seal and prime the fluid pump.

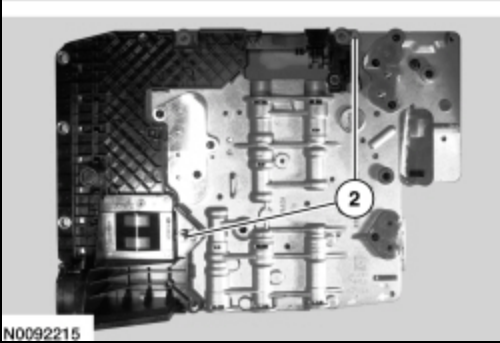
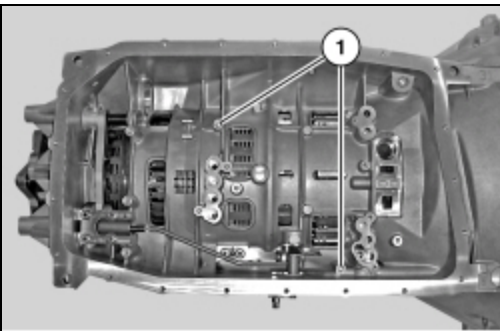


64. Install the thermal bypass valve into the case.



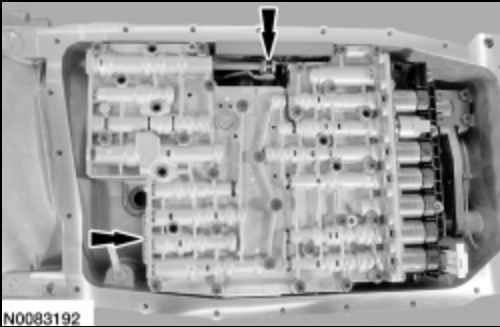
65. Align the guide pins in the alignment holes in the transmission case and position the mechatronic assembly in place.

1. Alignment holes
2. Guide pins



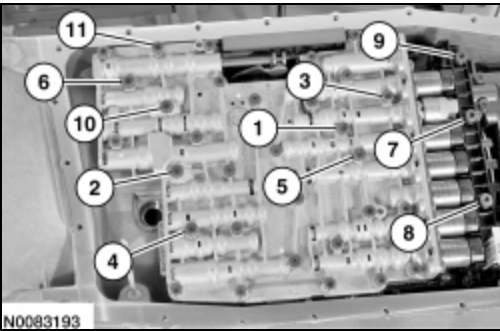
66. **NOTE:** *The mechatronic assembly will not lay flush on the case, this is a normal condition with the rubber feed tubes, the bolts will pull the mechatronic assembly down.*

Slightly lift the mechatronic assembly and align the manual valve in the manual valve linkage and position the mechatronic in place.

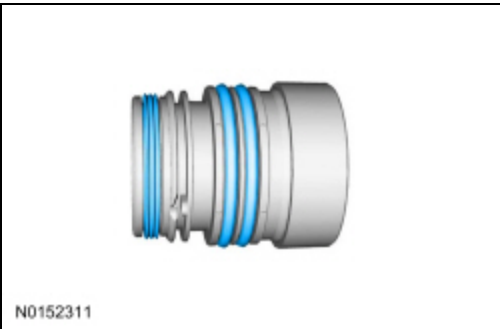


67. Install the 11 mechatronic assembly bolts and tighten in the sequence shown.

- Tighten to 8 Nm (71 lb-in).



68. If removed, install the bulkhead connector sleeve seals.



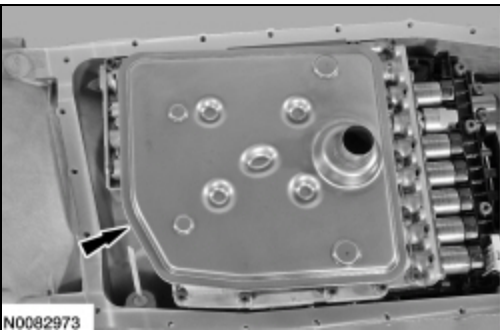
69. With the release tab up and unlocked, push the outer shell of the bulkhead connector sleeve into the transmission. Make sure the bulkhead connector sleeve is fully seated into the molded leadframe.



70. Press down on the tab and lock the outer shell of the bulkhead connector sleeve in place. Make sure the locking tab is securely locked.

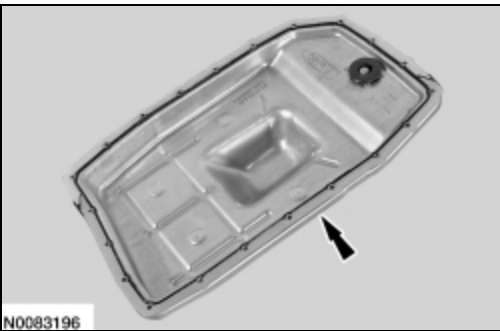


71. Install a new transmission fluid filter.



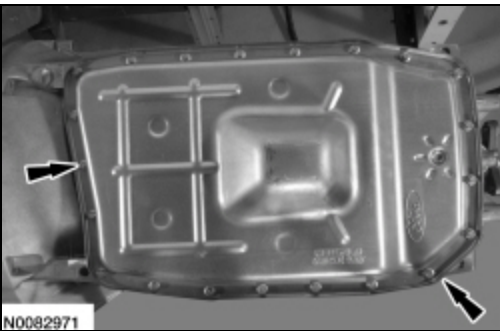
72. **NOTE:** *The fluid pan gasket can be reused if not damaged.*

Install a new transmission fluid pan gasket if required.



73. Install the transmission fluid pan and transmission fluid pan bolts. Tighten in a crisscross pattern.

- Tighten to 11 Nm (97 lb-in).



74. Using the Torque Converter Handle, install the torque converter.

